

Substitution: Reading an Integral Backwards Through the Chain Rule

Answer Key

AP Calculus AB/BC

Quick Answers

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|---|---|
| 1. $\frac{(x^2+7)^6}{6}$ | 4. $\sqrt{x^2+9}$ |
| 2. $\frac{e^{x^3}}{3}$ | 5. (a) upper u-limit = 1, (b) value = $\frac{1}{2}$ |
| 3. (a) lower u-limit = 1, (b) upper u-limit = 2, (c) value = $\frac{15}{4}$ | 6. (a) antiderivative = $\frac{\ln(x)^4}{4}$, — |
| | 7. (a) upper u-limit = 1, (b) value = $\frac{1}{2}$ |
| | 8. (a) value on $[0, 2]$ = $\frac{1-e^{-4}}{2}$, (b) limit = $\frac{1}{2}$, — |
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What is verified

13 of 15 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-6.D – problems 1, 2, 3, 4, 5, 6, 7

FUN-6.E – problem 8

Difficulty 1–5 of 5 across 15 tagged checks.

1. $\int 2x(x^2 + 7)^5 dx$, table supplied.

Solution. With $u = x^2 + 7$ and $du = 2x dx$ the integral becomes $\int u^5 du = \frac{u^6}{6} + C$.
Converting back to x :

$$\boxed{\frac{(x^2 + 7)^6}{6} + C}$$

Check by differentiating: $6 \cdot \frac{(x^2 + 7)^5}{6} \cdot 2x$ returns the integrand, which is what substitution promises.

2. $\int x^2 e^{x^3} dx$.

Solution. $u = x^3$, $du = 3x^2 dx$, so $x^2 dx = \frac{1}{3} du$:

$$\frac{1}{3} \int e^u du = \frac{1}{3} e^u + C.$$

$$\boxed{\frac{e^{x^3}}{3} + C}$$

3. $\int_0^1 2x(x^2 + 1)^3 dx$, $u = x^2 + 1$. (a) Lower u -limit. (b) Upper u -limit. (c) Value.

Solution (a). $u = 0^2 + 1$.

$$\boxed{u = 1}$$

Solution (b). $u = 1^2 + 1$.

$$\boxed{u = 2}$$

Solution (c). With $du = 2x dx$ the whole integral converts:

$$\int_1^2 u^3 du = \left[\frac{u^4}{4} \right]_1^2 = \frac{16 - 1}{4}.$$

$$\boxed{\frac{15}{4}}$$

Note the limits moved with the variable — nothing is converted back to x .

4. $\int \frac{x}{\sqrt{x^2 + 9}} dx$.

Solution. $u = x^2 + 9$, $du = 2x dx$, so $x dx = \frac{1}{2} du$:

$$\frac{1}{2} \int u^{-1/2} du = \frac{1}{2} \cdot 2u^{1/2} + C = \sqrt{u} + C.$$

$$\boxed{\sqrt{x^2 + 9} + C}$$

5. $\int_0^{\pi/4} \tan x \sec^2 x dx$, $u = \tan x$. (a) Upper u -limit. (b) Value.

Solution (a). $\tan \frac{\pi}{4}$.

$$\boxed{u = 1}$$

Solution (b). $du = \sec^2 x dx$, and the lower limit becomes $\tan 0 = 0$:

$$\int_0^1 u du = \left[\frac{u^2}{2} \right]_0^1.$$

$$\boxed{\frac{1}{2}}$$

6. $\int \frac{(\ln x)^3}{x} dx$. (a) Evaluate. (b) Why the $1/x$ matters.

Solution (a). $u = \ln x$, $du = \frac{dx}{x}$:

$$\int u^3 du = \frac{u^4}{4} + C.$$

$$\boxed{\frac{(\ln x)^4}{4} + C}$$

Solution (b) — instructor-judged. The substitution $u = \ln x$ forces $du = \frac{dx}{x}$, and the integrand already carries that $\frac{1}{x}$, so the differential is consumed exactly. Without it, an unconverted x would be left over that cannot be written in terms of u , and the substitution would stall. Full credit needs the $du = dx/x$ step, not just the remark that it works.

7. $\int_1^e \frac{\ln x}{x} dx$, $u = \ln x$. (a) Upper u -limit. (b) Value.

Solution (a). $\ln e$.

$$\boxed{u = 1}$$

Solution (b). The lower limit becomes $\ln 1 = 0$, and $du = \frac{dx}{x}$:

$$\int_0^1 u du = \left[\frac{u^2}{2} \right]_0^1.$$

$$\boxed{\frac{1}{2}}$$

8. $\int_0^b x e^{-x^2} dx$, $u = x^2$. (a) $b = 2$. (b) The limit. (c) What it means.

Solution (a). $du = 2x dx$, so $x dx = \frac{1}{2} du$, and u runs from 0 to b^2 :

$$\frac{1}{2} \int_0^{b^2} e^{-u} du = \frac{1}{2} (1 - e^{-b^2}).$$

At $b = 2$ this is

$$\boxed{\frac{1 - e^{-4}}{2}}$$

Solution (b). As $b \rightarrow \infty$, $e^{-b^2} \rightarrow 0$:

$$\boxed{\frac{1}{2}}$$

Solution (c) — instructor-judged. The region under $y = x e^{-x^2}$ stretches forever to the right, yet its total area settles on $\frac{1}{2}$: the tail contributes less and less because e^{-b^2} collapses to zero. The improper integral therefore *converges*, and its value is one half. Full credit needs the word *converges* or an equivalent statement that the area approaches a finite value rather than growing without bound.